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Q1 :

```
import java.util.*;
public class QueueDemo {
    int stack1[] = new int[10];
    int stack2[] = new int[10];
    int top1=-1;
    int front1=-1;
    int top2=-1;
    int sizeq;

    public void enqueue(int a) {
        if(top1==stack1.length-1) {
            System.out.println("Queue Overflow");
            return;
        }

        if(front1==-1) {
            front1 = 0;
        }
        top1++;
        stack1[top1]=a;
        sizeq++;
    }

    public void dequeue() {
        if(top1==-1) {
            System.out.println("EMPTY");
            return;
        }

        int temp = stack1[front1];
        front1++;
        sizeq--;

        top2++;
        stack2[top2]=temp;
        System.out.println(stack2[top2]);
        top2--;
    }

    public void peek() {
        if(top1==-1) {
            System.out.println("EMPTY");
            return;
        }
        System.out.println(stack1[front1]);
    }

    public void size() {
        System.out.println(sizeq);
    }
}
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    String choice;

    QueueDemo q = new QueueDemo();

    do {
        System.out.println("We are implementing queue");
        System.out.println("E : enqueue");
        System.out.println("D : dequeue");
        System.out.println("P : peek front value");
        System.out.println("S : size");
        System.out.println("X : exit");

        System.out.println("Enter your choice");
        choice = sc.next();
        sc.nextLine();

        switch(choice) {
            case "E" : {
                System.out.println("Enter element to be added");
                int a = sc.nextInt();
                q.enqueue(a);
                break;
            }

            case "D" : {
                q.dequeue();
                break;
            }

            case "P" : {
                q.peek();
                break;
            }

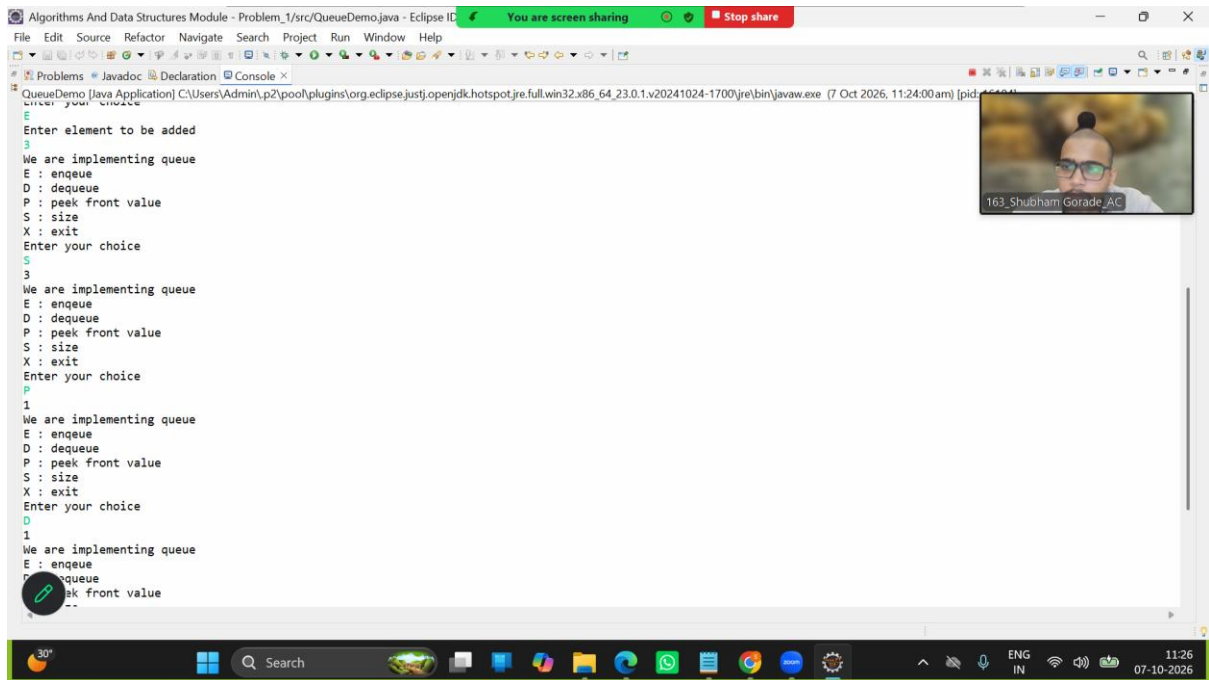
            case "S" : {
                q.size();
                break;
            }

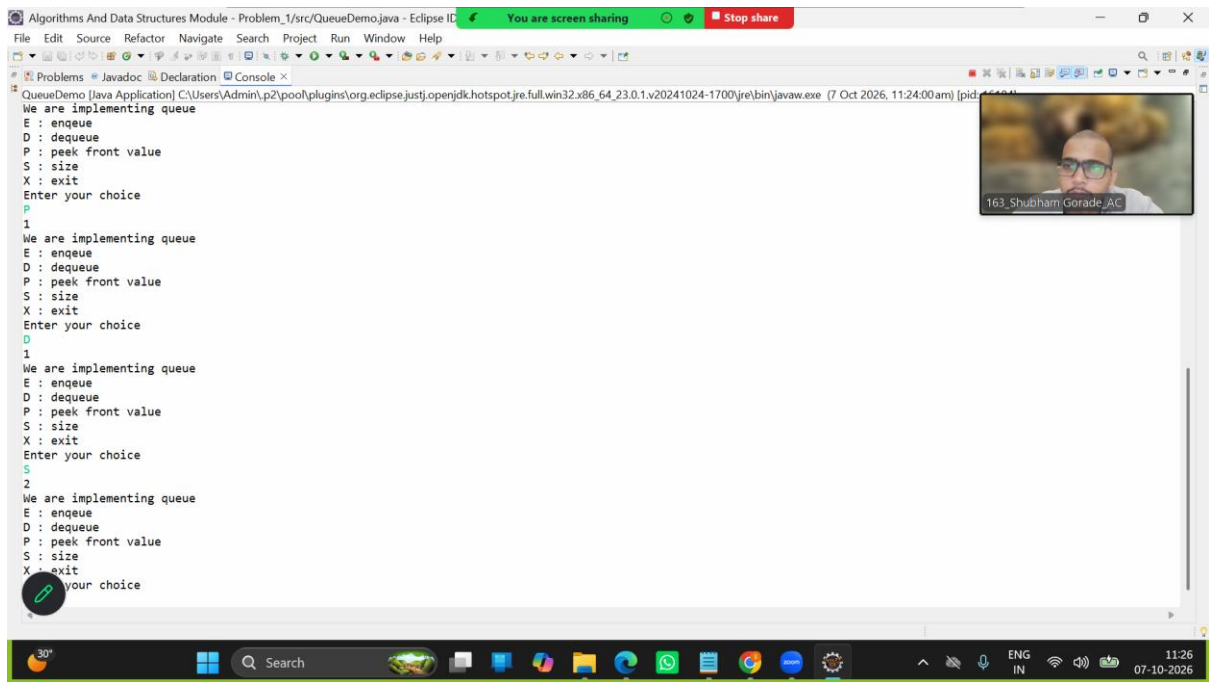
            case "X" : {
                System.out.println("Exiting the program");
                break;
            }

            default : {
                System.out.println("Enter correct choice");
                break;
            }
        }
    }while(choice!="X");
}

```

Output :





Q2:

```
import java.util.*;
public class LinkedListDuplicates {
    class Node {
        int data;
        Node next;

        Node(int data) {
            this.data = data;
        }
    }

    Node head;

    public void add(int d) {
        Node n1 = new Node(d);
        if(head==null) {
            head = n1;
            System.out.println("Added in list"+d);
            return;
        }

        n1.next = head;
        head = n1;
        System.out.println("Added in list"+d);
    }

    public void printDuplicates() {
        System.out.println("Execute");
        Node temp1 = head;
        Node temp2 = head.next;
        for(int i=0; temp1.next!=null; i++) {
            int count = 0;
```

```

        for(int j=0; temp2.next!=null; j++) {
            if(temp1.data==temp2.data) {
                count++;
            }
            temp2=temp2.next;
        }
        if(count==0) {
            System.out.println("No Duplicates");
        }
        else {
            System.out.println("Duplicates"+ temp1.data + " "+ count
+" "+"times");
        }
        temp1=temp1.next;
    }

}

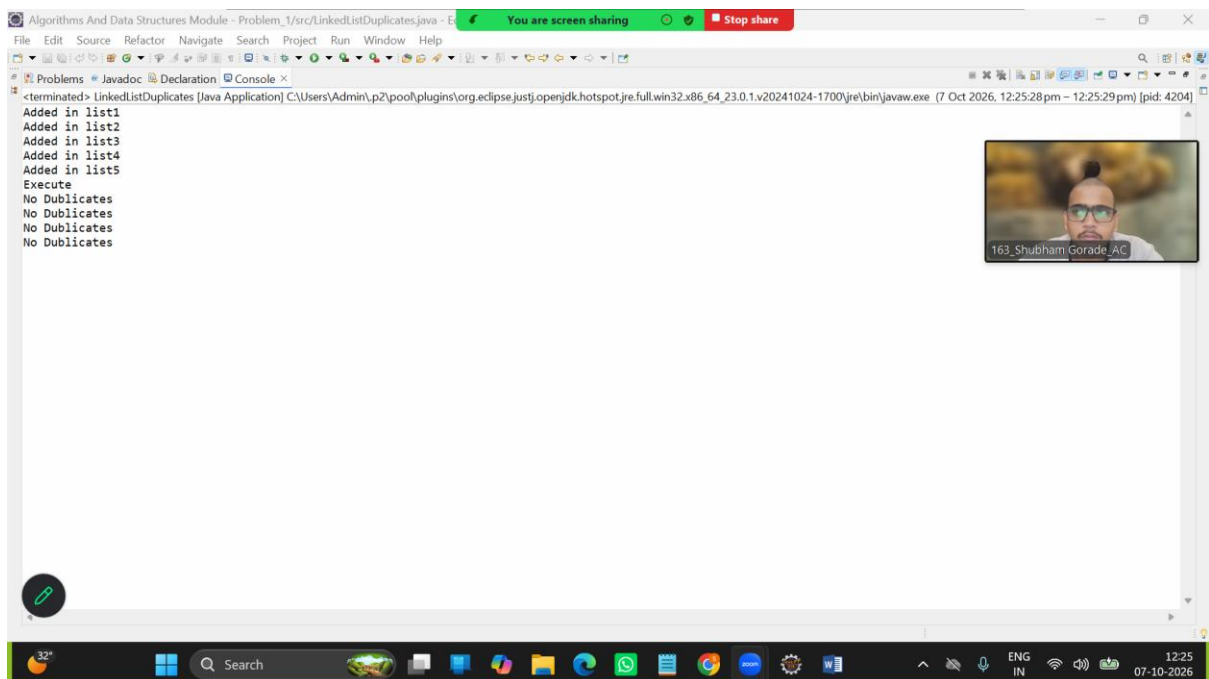
public static void main(String args[]) {

    LinkedListDuplicates l = new LinkedListDuplicates();
    l.add(1);
    l.add(2);
    l.add(3);
    l.add(4);
    l.add(5);

    l.printDuplicates();
}
}

```

Output :



The screenshot shows an IDE window titled "Algorithms And Data Structures Module - Problem_1/src/LinkedListDuplicates.java". The console output is as follows:

```

<terminated> LinkedListDuplicates (Java Application) C:\Users\Admin\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_23.0.1.v20241024-1700\jre\bin\javaw.exe (7 Oct 2026, 12:25:28 pm - 12:25:29 pm) [pid: 4204]
Added in list1
Added in list2
Added in list3
Added in list4
Added in list5
Execute
No Duplicates
No Duplicates
No Duplicates
No Duplicates

```

A video call window is visible in the bottom right corner of the IDE, showing a person with a beard and glasses, identified as "163_Shubham Gorade.AC".

Q3:

```
import java.util.*;
public class BSTInorder {
    class Node {
        int data;
        Node left;
        Node right;

        Node(int data) {
            this.data = data;
        }

        public void add() {
            Node n1 = new Node(int d);
        }
        public static void main(String args[]) {

        }
    }
}
```

Output :

